Can an Action Potential Magnetically Drive Mitochondrial ATP Synthesis?

A Geometry-Aware Feasibility, Identifiability, and Discriminating-Experiment Analysis of a 2022 Self-Aware Networks Hypothesis

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Abstract

Neuronal firing creates an immediate energy bill. Ionic gradients must be restored, vesicles recycled, proteins maintained, and synapses supplied. The established account connects that bill to local mitochondrial response through substrate demand, calcium signaling, transport, and metabolic control. Self-Aware Networks (SAN) notes publicly fixed in June 2022 proposed a more specific addition: the electromagnetic field accompanying neuronal firing could alter mitochondrial flavin chemistry, increase ATP production, and thereby influence plasticity. This paper asks what that proposal would have to mean physically and how it can be distinguished from ordinary activity-to-metabolism coupling.

Four routes are kept separate: established biochemical demand coupling; calcium-mediated mitochondrial control; ordinary Faraday induction; and chemically specific radical-pair transduction. A unique organelle exposure cannot be reconstructed from action-potential voltage alone. It requires compartment-resolved currents, intra- and extracellular electrical properties, return-current geometry, mitochondrial position and orientation, crista geometry, and the local background field. We therefore compute transparent geometry-aware screening bounds rather than claiming a complete neuronal field solution. A 10 nA reference current in a uniform 0.5 micrometre radius cylinder gives 2 nT at the declared organelle position and a closed-path circulation electromotive force of 6.28×10^{-19} V. A deliberately favorable 1 microampere internal case gives 0.28 microtesla and 1.27×10^{-15} V. An ideal-line near-source ceiling gives 1.54 microtesla and 1.01×10^{-14} V. Even the ceiling supplies less than 4×10^{-13} of thermal energy per elementary charge. Ordinary induction is therefore quantitatively negligible in these favorable geometries.

That no-go result does not decide the radical-pair route, which changes chemical branching rather than supplying bulk activation energy. The relevant electron-transfer-flavoprotein pilot compared 20 nT with 50 microtesla but found no statistically significant difference in either hydrogen peroxide or superoxide production. A separate redox-flavoenzyme study likewise found no magnetic response under its tested conditions. Thus neither source demonstrates neuronal magnetic control of ATP. The radical-pair branch remains a conditional hypothesis requiring a nominated molecular pair, compatible lifetime and kinetics, measured endogenous waveform, amplification, and an ATP consequence. A staged purified-protein, isolated-mitochondrion, and neuron-level cancellation-and-rescue program is specified to decide it. The result is a typed verdict: activity-dependent mitochondrial support of plasticity is established; direct induction fails the scale test; flavin radical-pair mediation remains unvalidated but experimentally decidable.

biomagnetism; synaptic plasticity; calcium; identifiability; feasibility analysis.

Keywords: Self-Aware Networks; mitochondria; action potential; ATP; flavin; radical pair;

1. The ordinary problem comes first

A neuron does not fire for free. During and after electrical activity, ion pumps restore sodium, potassium, and calcium gradients; synaptic vesicles are recycled; transmitter is repackaged; cytoskeletal and membrane machinery are maintained; and local plasticity can require protein synthesis and structural change. Synaptic transmission is a major energy consumer, and experimental work shows that local ATP synthesis is required to sustain it [8]. Mitochondrial calcium handling can regulate neuronal metabolism and long-term memory across species [9], while learning and plasticity signals can reorganize ATP synthase in synaptic mitochondria [10].

The first causal spine is therefore uncontroversial in its broad form:

$$S(t) \longrightarrow \{D_{\text{ion}}, C_{\text{mito}}, U_{\text{sub}}, T_{\text{org}}\} \longrightarrow M(t) \longrightarrow \{A_{\text{ATP}}, P_{\text{plasticity}}\}. \tag{1}$$

Here D_{ion} is ionic-restoration demand, C_{mito} mitochondrial calcium, U_{sub} substrate use, T_{org} organelle transport or reorganization, and $M(t)$ the resulting mitochondrial response. Equation (1) does not say that ATP is memory, that every firing event creates a synapse, or that one organelle decides what the neuron learns. It says that activity-dependent energetic state helps determine which transmission and plasticity processes can be sustained.

The unresolved question is narrower. Does the electromagnetic field accompanying the firing event contribute information or control to the mitochondrial response after ordinary demand and calcium routes have been accounted for? The question is worth asking because the alternatives make different predictions. If metabolism responds only to demand and biochemical signaling, cancelling a local magnetic waveform while preserving voltage, current load, calcium, temperature, and substrates should not change the ATP response. If a chemically specific magnetic sensor contributes, then cancellation should remove a reproducible residual and waveform-matched magnetic rescue should restore it.

Only after that problem and discriminator are clear is a name useful. The 2022 SAN corpus called this candidate joined operation a firing-field-to-mitochondrial-flavin-to-ATP learning mechanism. In compact notation:

$$S(t) \longrightarrow B_{\text{AP}}(\mathbf{x}, t) \longrightarrow F_{\text{mito}}(t) \longrightarrow A_{\text{ATP}}(t) \longrightarrow P_{\text{plasticity}}(t), \tag{2}$$

where $S(t)$ is a firing event, B_{AP} its local magnetic perturbation, F_{mito} a candidate field-sensitive mitochondrial chemical state, A_{ATP} ATP availability or production capacity, and $P_{\text{plasticity}}$ a downstream change in the capacity for transmission, protein synthesis, receptor organization, structural growth, or learning.

2. The dated SAN proposition and its recovered intent

The proposition was not reconstructed from later terminology. Public Git records from June 2022 already contain its component operations. On June 8, `a0607z` joined neuronal firing, electromagnetic excitation, mitochondrial ATP, and faster learning; `a0613z` represented a state transition from soma firing through mitochondrial activity and protein synthesis to receptor growth [1,2]. The original uploaded forms of `a0101z` and `a0104z` were also fixed on June 8. They connected flavins, electron-transfer-flavoprotein chemistry, ATP, growth, radical pairs, singlet / triplet states, field strength, and frequency windows [3,4]. `a0245z` stated the full axonal-field-to-ATP-to-synaptic-growth chain on June 11 [5]. `a0617z` placed the proposal within a tonic / phasic field architecture and described ATP as a cell-local reinforcement constraint [6].

Earlier recordings show the conceptual approach developing before the complete operator. An April 17, 2021 Google Recorder source treats activity as an ATP supply-and-demand problem and warns that stimulation can increase demand without increasing supply. An October 11 recording relates pattern completion to electrical, magnetic, and chemical momentum. A November 1 recording states that greater stimulation requires greater ATP and proposes several possible physical routes from activity to growth. Those recordings are legitimate ancestry for the problem formulation. They are not silently promoted into the complete June 2022 flavin operator.

Some early notes use compressed or physiologically imprecise language. This successor recovers the strongest source-consistent meaning without rewriting the historical record:

- "mitochondria fire" or "release ATP" is read as mitochondria changing ATP synthesis, export, or available metabolic capacity, not as an organelle action potential or secretory event;
- "electromagnetic wave" is separated into measured electric currents, their magnetic field, and possible mechanochemical effects rather than treated as one undifferentiated cause;
- a notation that placed a mitochondrion with a nucleus is treated as a diagrammatic location slip: mitochondria lie within the cell but outside the nucleus;
- "flavins are sensitive" becomes the testable claim that a particular flavin-containing reaction might host an appropriately organized radical pair, not that all flavins are magnetosensors;
- "reward" means a change in metabolic capacity that can alter the probability and persistence of future plasticity, not a subjective reward experienced by one cell;
- ATP is a permissive and regulatory resource, not the informational content of memory.

This intent-recovery rule is conservative in both directions. It does not preserve a likely typo as biology, and it does not erase a legitimate hypothesis merely because an early note compressed several causal levels into one sentence. The immutable source wording and the present operational formulation remain separately inspectable in the accompanying atlas.

3. Four routes that must not be collapsed

Let an observed activity-linked ATP response be decomposed for experimental purposes as

$$\Delta A_{\mathrm{ATP}} = \Delta A_{\mathrm{demand}} + \Delta A_{\mathrm{Ca}} + \Delta A_{\mathrm{ind}} + \Delta A_{\mathrm{RP}} + \Delta A_{\mathrm{other}}. \tag{3}$$

The terms are not assumed to be statistically independent. Equation (3) is a bookkeeping requirement for designing interventions.

3.1 Demand and ordinary metabolic control

Firing increases pump work, vesicle cycling, transmitter handling, and maintenance costs. Local ATP synthesis is required for sustained synaptic function [8]. This route needs no magnetic mediator. It predicts that matched electrical and synaptic workload should produce a metabolic response even when a putative local magnetic component is cancelled.

3.2 Calcium and signaling control

Activity changes cytosolic and mitochondrial calcium. Calcium uptake and efflux interact with metabolic enzymes, membrane potential, transport, and ATP production. Cross-species evidence now links mitochondrial calcium efflux to neuronal metabolism and long-term memory [9]. This route is also nonmagnetic in the relevant causal sense, even though every ionic current has an associated electromagnetic field.

3.3 Ordinary electromagnetic induction

A time-varying magnetic flux produces a circulation electric field around a closed path. It does not automatically produce a voltage "across a mitochondrion." A mitochondrial inner membrane, outer membrane, and crista network are not identified pickup-coil terminals. The relevant screening observable is therefore a closed-path electromotive force (EMF), followed by a comparison of the work per elementary charge with thermal scales. This route can be bounded directly.

3.4 Spin-selective radical-pair chemistry

A radical pair can, under particular molecular and kinetic conditions, alter singlet / triplet evolution and chemical product branching in response to magnetic field. The field need not provide bulk thermal activation energy. This route is therefore not refuted by a tiny induction voltage or a tiny single-spin Zeeman-to-thermal ratio. It requires a separate molecular test.

The central methodological rule follows: evidence for activity-dependent ATP does not establish a magnetic mediator; failure of ordinary induction does not refute every radical-pair mechanism; and an in-vitro radical-pair response would not by itself establish an endogenous neuronal route.

4. Why the exact organelle field is not yet identifiable

Hodgkin and Huxley provided a quantitative membrane-current model for the action potential [11], but membrane voltage alone does not determine the magnetic field at a nearby mitochondrion. Neuronal magnetic fields depend strongly on axial current and also on extracellular return currents and the electrical properties of the media [12,13]. A field solution at an organelle requires at least

$$\boldsymbol{\theta}_{\text{field}} = \{\mathbf{J}_{\text{axial}}(\mathbf{x},t), \mathbf{J}_{\text{mem}}(\mathbf{x},t), \sigma_i(\omega), \sigma_e(\omega), \mathcal{G}_{\text{return}}, \mathcal{G}_{\text{cell}}, \mathcal{G}_{\text{mito}}, \mathbf{B}_0\}, \tag{4}$$

where the two current fields, intracellular and extracellular conductivity or impedance, return-current geometry, cell morphology, mitochondrial position and orientation, crista geometry, and local static background field must be known or bounded.

Current this paper inputs do not identify all of $\boldsymbol{\theta}_{\text{field}}$. It would be false precision to publish a unique "field across the mitochondrion." The correct response is not to stop calculating; it is to calculate geometry-aware screening cases whose assumptions and direction of conservatism are explicit.

Mitochondrial dimensions provide context for those cases. Three-dimensional electron microscopy of mouse cerebellar axonal and presynaptic mitochondria reports example lengths near 0.5-0.7 micrometres and widths near 0.20-0.24 micrometres, with nontrivial crista morphology [14]. Those numbers motivate submicrometre loop scales; they are not universal organelle dimensions or evidence that a biological membrane forms an electrical pickup loop.

5. Geometry-aware field and induction bounds

5.1 Internal current-cylinder cases

For an infinite cylinder of radius R carrying a uniformly distributed axial current I, the azimuthal field inside is

$$B_\phi(r) = \frac{\mu_0 I r}{2\pi R^2}, \qquad 0 \le r \le R. \tag{5}$$

Consider a circular test loop of radius a, lying in a radial-axial plane, centered at radial offset d, and wholly inside the cylinder so that $d + a < R$. Integrating the field over that surface gives

$$\Phi_{\text{in}} = \frac{\mu_0 I d a^2}{2R^2}. \tag{6}$$

If the current changes over rise time Δt, the magnitude of the circulation EMF is

$$|\mathcal{E}_{\text{in}}| = \frac{\mu_0 a^2 d}{2R^2} \left| \frac{\Delta I}{\Delta t} \right|. \tag{7}$$

The reference case sets $I = 10$ nA, $R = 0.50$ micrometres, $a = 0.10$ micrometres, $d = 0.25$ micrometres, and $\Delta t = 100$ microseconds. It gives a 2 nT center field and 6.28×10^{-19} V circulation EMF.

The generous internal case sets $I = 1$ microampere, the same cylinder radius, $a = 0.12$ micrometres, $d = 0.35$ micrometres, and $\Delta t = 10$ microseconds. It gives 0.28 microtesla and 1.27×10^{-15} V. This parameter set is a deliberately favorable ceiling, not a claim about a typical neuron.

5.2 External ideal-line ceiling

An ideal infinite line current gives

$$B_\phi(r) = \frac{\mu_0 I}{2\pi r}. \tag{8}$$

For a circular radial-axial loop of radius a centered a distance $d > a$ from the line, exact surface integration gives

$$\Phi_{\text{line}} = \mu_0 I \left(d - \sqrt{d^2 - a^2} \right), \tag{9}$$

and therefore

$$|\mathcal{E}_{\text{line}}| = \mu_0 \left(d - \sqrt{d^2 - a^2} \right) \left| \frac{\Delta I}{\Delta t} \right|. \tag{10}$$

Setting $I = 1$ microampere, $a = 0.12$ micrometres, $d = 0.13$ micrometres, and $\Delta t = 10$ microseconds gives a 1.54 microtesla center field and 1.01×10^{-14} V. The near-singular line geometry is more favorable to the hypothesis than a finite distributed source and is used only as a screening ceiling.

Declared case	Geometry	Center field	Closed-path EMF	Field / 50 microtesla
Reference internal	uniform current cylinder	2 nT	6.28×10^{-19} V	4.0×10^{-5}
Generous internal	uniform current cylinder	0.28 microtesla	1.27×10^{-15} V	5.6×10^{-3}
Near-line ceiling	ideal line current	1.54 microtesla	1.01×10^{-14} V	3.08×10^{-2}

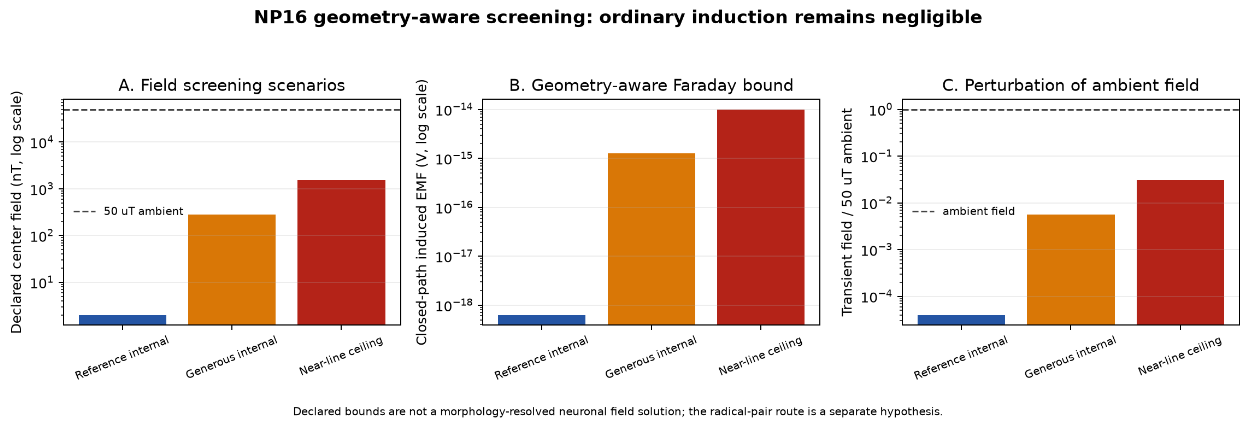

Figure 1: Geometry-aware field and induction screening

5.3 What the EMF means

The EMF is a closed-path circulation. It is not the mitochondrial inner-membrane potential, and dividing it by a 150 mV membrane-potential reference compares unlike observables. A safer energetic screen asks how much work one elementary charge could receive from the circulation:

$$\eta_{\text{ind}} = \frac{e|\mathcal{E}|}{k_B T}. \tag{11}$$

At 310 K, the three cases give values far below 10^{-9}; even the near-line ceiling is below 4×10^{-13}. No plausible orientation correction converts that circulation into a thermal-scale or mitochondrial-membrane-scale actuator. Ordinary Faraday induction is therefore negligible in all three declared cases.

This conclusion is a scale result, not a claim that neuronal magnetic fields do not exist. Wikswo, Barach, and Freeman measured a 0.12 nT compound frog sciatic-nerve field at 1.3 mm [15]. That is an aggregate preparation measurement, not a single-neuron organelle exposure, and it cannot be projected inward by a naive $1/r$ rule. Later volume-conductor work explicitly treats source and return currents [12,13].

6. The radical-pair route survives the induction no-go only conditionally

A coarse single-electron energetic comparison is

$$\eta_Z = \frac{\mu_B B}{k_B T}. \tag{12}$$

For fields in the nT-to-low-microtesla cases above, η_Z is thermally tiny. That excludes a story in which the field simply supplies a generic activation energy. It is not a theorem against radical-pair chemistry. Radical pairs can respond through coherent spin evolution, hyperfine coupling, relaxation, recombination, molecular orientation, and product branching.

The 2022 `a0104z` note is important precisely because it named this more specific route: singlet / triplet states, flavins, electron-transfer flavoprotein, field magnitude, and frequency windows [4]. Its viability can be written as a conjunction:

$$H_{\mathrm{RP}} = H_B \wedge H_\tau \wedge H_{\mathrm{geom}} \wedge H_{\mathrm{chem}} \wedge H_{\mathrm{gain}} \wedge H_{\mathrm{ATP}}, \tag{13}$$

where:

- H_B: the endogenous field amplitude, vector, waveform, and background are compatible;
- H_τ: the radical-pair lifetime and relaxation permit a response;
- H_{geom}: molecular geometry and orientational statistics preserve sensitivity;
- H_{chem}: the field changes a consequential product branch;
- H_{gain}: the molecular change survives noise and reaches cellular scale;
- H_{ATP}: the change alters ATP synthesis or availability rather than only a small reactive-oxygen-species ratio.

No recovered source establishes all six conditions in a firing neuron.

6.1 What the electron-transfer-flavoprotein pilot actually found

Austvold and colleagues tested recombinant human electron-transfer flavoprotein under approximately 20 nT and 50 microtesla static-field conditions [16]. Hydrogen peroxide was 2.1 ± 0.3 versus 2.4 ± 0.2 micromolar, a numerical difference of roughly 13%, but $p = 0.36$. Superoxide was 10.6 ± 1.4 versus 9.6 ± 0.8 micromolar, a numerical difference of roughly 10%, but $p = 0.29$. Both outcomes were non-significant. The study is a small, mechanistically motivated pilot, not positive evidence of an effect.

Its field contrast also differs sharply from the neuronal hypothesis. It compared near-removal of the ambient static field with an approximately geomagnetic condition, a 2,500-fold field difference. A 2 nT transient superposed on 50 microtesla changes field magnitude by only 4×10^{-5} in the favorable orientation. Even the declared near-line ceiling is about 3.1% of 50 microtesla. The pilot therefore neither validates nor directly tests an action-potential-scale perturbation.

Messiha and colleagues tested multiple redox flavoenzymes and a model flavin-nicotinamide system and found no magnetic response under their conditions [17]. Their result likewise does not prove that no radical-pair system can respond; it establishes that "contains flavin" is not a sufficient mechanism. Together the two sources set the proper evidence ceiling: the proposed route is chemically specific, currently unvalidated, and testable.

7. The strongest surviving SAN contribution

The most durable part of the 2022 proposal is not the ordinary-induction branch. It is the decision to place activity-dependent mitochondrial state inside the learning operator rather than outside it as a passive utility. The neuron's recent activity changes its energetic requirements and signaling state; those changes influence whether transmission, structural maintenance, and plasticity can continue. In that bounded sense, metabolic state helps determine the future accessibility of learned routes.

The mechanism can now be written without collapsing evidence levels:

$$S \longrightarrow \{D, C, T, U\} \longrightarrow A_{\mathrm{ATP}} \longrightarrow P \qquad \text{established}, \tag{14}$$

$$\begin{gathered} S \longrightarrow B_{\mathrm{AP}} \longrightarrow \mathcal{E}_{\mathrm{ind}} \longrightarrow A_{\mathrm{ATP}} \\ \text{negligible in declared bounds,} \end{gathered} \tag{15}$$

$$\begin{gathered} S \longrightarrow B_{\mathrm{AP}} \longrightarrow R_F \longrightarrow Q_{\mathrm{chem}} \longrightarrow A_{\mathrm{ATP}} \\ \text{conditional, unvalidated.} \end{gathered} \tag{16}$$

In Equation (14), D, C, T, U abbreviate demand, calcium, transport, and substrate use. In Equation (16), R_F is the nominated flavin radical pair and Q_{chem} its changed chemical product branch. This typed result strengthens rather than dissolves the original research program. It identifies what is already supported, what fails quantitatively, and what exact residual remains worth testing.

8. A staged experiment that can decide the residual

The proposed test should not begin by stimulating a neuron and measuring ATP alone. Such a result would be dominated by ordinary workload and calcium. The program needs three stages.

8.1 Stage I: purified molecular system

Replicate the electron-transfer-flavoprotein assay with adequate power, blinded analysis, and prespecified outcomes. Sweep static background field, transient amplitude, waveform, orientation, rise time, and repetition rate. Include the exact nT-to-low-microtesla transients predicted or measured near neurons, not only near-zero-versus-geomagnetic contrasts. Measure radical-pair or spin-sensitive observables where possible, product yields, ROS partitioning, and reaction kinetics. Estimate a complete dose-response surface rather than declaring a single condition "magnetic."

Stage I succeeds only if a reproducible field-dependent chemical branch is found and connected to a specific molecular pair and kinetic model. A null result with adequate sensitivity rejects that tested molecular branch.

8.2 Stage II: isolated mitochondria

Expose isolated neuronal mitochondria or a controlled mitochondrial preparation to the verified waveform while matching temperature, vibration, oxygenation, pH, substrates, and mechanical conditions. Measure membrane potential, oxygen consumption, ATP synthesis, ROS, flavin redox state, and calcium. Perturb the nominated flavoprotein or radical-pair lifetime while preserving general mitochondrial viability.

Stage II succeeds only if the field response reaches ATP or another prespecified mitochondrial endpoint and follows the molecular perturbation predicted at Stage I.

8.3 Stage III: neuron-level cancellation and rescue

In cultured neurons or an ex-vivo preparation, acquire synchronized measurements of:

- membrane voltage and compartment-resolved current;
- vector magnetic field at calibrated subcellular positions;
- mitochondrial calcium and membrane potential;
- a specific flavoprotein redox or spin-sensitive reporter;
- oxygen consumption or respiratory flux;
- compartment-targeted ATP;
- temperature, motion, pH, ROS, substrate supply, and spike workload.

Use four core conditions:

1. ordinary firing;

2. matched firing with local cancellation of the measured candidate magnetic waveform at the mitochondrial region;
3. cancellation plus waveform-matched magnetic rescue;
4. sham actuation with matched heat, vibration, and instrumentation.

Spike count, voltage trajectory, calcium transient, pump load, substrate delivery, and temperature must fall within preregistered equivalence bounds across the relevant conditions. Define the magnetic residual as a causal contrast:

$$R_B = \Delta A_{\mathrm{ATP}}^{\mathrm{ordinary}} - \Delta A_{\mathrm{ATP}}^{\mathrm{cancelled}}, \tag{17}$$

and the rescue contrast as

$$R_{\mathrm{rescue}} = \Delta A_{\mathrm{ATP}}^{\mathrm{rescued}} - \Delta A_{\mathrm{ATP}}^{\mathrm{cancelled}}. \tag{18}$$

The magnetic branch is supported only if:

- the endogenous vector waveform is measured rather than inferred from voltage alone;
- cancellation changes the endpoint while electrical workload, calcium, temperature, and substrates remain equivalent;
- waveform-matched rescue restores the effect;
- flavoprotein or radical-pair perturbation changes the effect in the predicted direction;
- the result replicates with a prespecified effect size and independent analysis.

If cancellation and rescue fail at the measured endogenous amplitude, the tested magnetic branch is rejected while Equation (14), the established activity-to-metabolism relation, remains intact.

9. Scope, limitations, and falsifiers

The cylinder and line-current cases are analytic screens, not cable-equation, volume-conductor, or organelle-resolved simulations. Current magnitude, geometry, organelle position, and rise time are declared scenarios. Mitochondrial dimensions are informed by one quantified preparation and do not define every neuron. A full calculation requires the parameter set in Equation (4).

The direct-induction result is nevertheless robust within its stated target. Its most favorable declared geometry produces only 1.01×10^{-14} V of closed-path EMF and less than $4 \times 10^{-13} k_B T$ of work per elementary charge. The calculation does not compare that EMF as though it were a transmembrane voltage.

The radical-pair branch remains less identified. Response can be nonmonotonic, orientation dependent, and chemically specific. ROS partitioning is not ATP production; ATP availability is not memory content; mitochondrial support is not proof that a single metabolic route selected a learned representation.

The decisive falsifiers are correspondingly typed:

- **ordinary induction:** already rejected for the declared biological screening range unless a measured geometry and current exceed the bound by many orders of magnitude;
- **nonspecific magnetic energy:** rejected by the thermal-scale comparison;
- **the nominated radical pair:** rejected if adequately powered molecular tests find no response at the measured neuronal waveform, or if cell-level cancellation and rescue fail;
- **the broader metabolic-learning link:** not rejected by failure of either magnetic route and remains supported through demand, calcium, transport, and substrate mechanisms.

All model inputs, formulas, result values, and checks are available in model/run_geometry_aware_field_bounds.py and model/geometry-aware-field-bounds.json. Passing those checks establishes arithmetic consistency, not biological truth.

10. Conclusion

The 2022 SAN corpus posed a specific question ahead of this calculation: could the field associated with neuronal firing alter mitochondrial flavin chemistry, increase ATP availability, and thereby change plasticity capacity? The question contains one well-supported architecture and two very different magnetic implementations.

Activity-dependent mitochondrial support of transmission, plasticity, and memory is the established spine. Geometry-aware bounds exclude ordinary Faraday induction as the relevant actuator at the declared scales. The correction is stronger than a generic "the field is small": a unique organelle field is not identifiable from action-potential voltage alone, a mitochondrial membrane is not an identified pickup coil, and the favorable closed-path EMF remains negligible.

The flavin radical-pair version cannot be dismissed by that induction calculation, but it has not been demonstrated. The available electron-transfer-flavoprotein pilot reported only non-significant numerical differences between 20 nT and 50 microtesla, while another redox-flavoenzyme study found no magnetic response. The surviving claim is therefore precise: a particular mitochondrial flavin-containing reaction might transduce a measured neuronal magnetic waveform through spin-selective product branching, but only if the field, lifetime, geometry, chemistry, amplification, and ATP-consequence conditions all hold.

That residual is scientifically useful because it can be decided. Molecular dose-response tests, isolated-mitochondrial experiments, and neuron-level field cancellation with matched magnetic rescue can separate it from ordinary workload and calcium. The final verdict is neither an all-or-nothing endorsement nor an all-or-nothing rejection. SAN's joined mitochondrial-learning problem remains fruitful; direct induction fails; the radical-pair mediator is an unvalidated, bounded, and falsifiable research hypothesis.

Authorship and development boundary

Micah Blumberg is the author and theory originator. AI tools assisted with source routing, calculation, manuscript development, and internal consistency review. The claims and evidence boundaries in this manuscript are intended for public preprint review; publication and repository releases remain decisions of the author.

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